

Cross-domain evaluation of ResNet50 and Swin-T for maize disease classification

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Repository: <https://github.com/a-proton/nepalcropnet>

Abstract

We trained ResNet50 and Swin Transformer (Swin-T) on the PlantVillage maize subset and tested both on PlantDoc, a smaller dataset of plant disease images from web search results. On PlantVillage, both models exceeded 97% test accuracy (Swin-T: 97.52 macro F1; ResNet50: 96.98). On PlantDoc, accuracy fell to 39.42% for Swin-T and 23.81% for ResNet50. Prototype-based few-shot classification with 20 in-domain support images reached 95.17%, close to the supervised result. Five PlantDoc support images per class raised cross-domain accuracy from 39.42% to 51.02%, still well below in-domain performance. Both models classified roughly one in five diseased PlantDoc images as healthy.

1. Introduction

PlantVillage (Hughes & Salathé, 2015) is the most common training dataset for plant disease classification. Models trained on it typically report over 95% accuracy, but PlantVillage images are taken in lab conditions: uniform lighting, plain backgrounds, single leaves at consistent angles. Field deployment looks different.

PlantDoc (Singh et al., 2020) was assembled from web search results to represent more realistic conditions. It has fewer images per class and substantially more visual variation. Comparing performance on the two datasets gives a measure of how transferable PlantVillage-trained models are.

This project addresses four questions:

1. How much does accuracy drop from PlantVillage to PlantDoc?
2. Does Swin-T transfer better than ResNet50?
3. Do prototype-based few-shot methods help close the gap with limited PlantDoc labels?
4. Do failures concentrate on safe mistakes (wrong disease) or unsafe ones (predicting health on diseased plants)?

2. Methodology

2.1 Datasets

Dataset	Conditions	Maize classes	Images
PlantVillage	Lab	Gray Leaf Spot, Common Rust, Northern Leaf Blight, Healthy	3,852
PlantDoc	Web search	Gray Leaf Spot, Common Rust, Northern Leaf Blight	378

PlantVillage was split 70/15/15 into train, validation, and test (stratified by class, seed=42).

PlantDoc was held out entirely; neither model was trained on any of it. PlantDoc has no Corn Healthy class, so cross-domain evaluation is over 3 classes.

2.2 Models

ResNet50. ImageNet V2 weights, final FC replaced with Dropout(0.3) $\rightarrow$ Linear(2048, 4). Trained 15 epochs with AdamW (lr=1e-4, weight_decay=0.05), cosine LR schedule, weighted random sampler, and standard augmentation (random crop, horizontal flip, $\pm 15^\circ$ rotation, color jitter).

Swin-T. ImageNet V1 weights, head replaced with Dropout(0.3) $\rightarrow$ Linear(768, 4). Two-stage training: 5 epochs with the backbone frozen and only the head training, then 10 epochs with the last Swin stage and final LayerNorm unfrozen at lr=1e-4. The two-stage approach is common for transformers on small datasets.

2.3 Few-shot evaluation

Following Snell et al. (2017), Swin-T (after Stage 2) is used as a frozen feature extractor producing 768-dimensional embeddings. Each episode samples N classes, K support images per class, and 15 query images per class. The prototype for each class is the mean of its support embeddings; queries are assigned to the nearest prototype. Reported numbers are mean $\pm$ 95% CI over 1,000 episodes per setting, for $K \in \{1, 5\}$ and both Euclidean (raw embeddings) and cosine (L2-normalized) distance.

3. Results

3.1 PlantVillage test set

Model	Test Acc	Macro F1	Gray	Rust	Blight	Healthy
ResNet50	97.92	96.98	91.89	100.0	96.03	100.0
Swin-T	98.27	97.52	93.42	100.0	96.64	100.0

Both models reached 100% F1 on Common Rust and Healthy. The 0.5-point macro F1 gap between Swin-T and ResNet50 comes entirely from Gray Leaf Spot, where Swin-T scored 93.42 to ResNet50's 91.89.

3.2 Few-shots on PlantVillage

Setting	Accuracy	95% CI	Labels
4-way 1-shot, cosine + L2-norm	88.94%	$\pm 0.37\%$	4
4-way 5-shot, Euclidean	95.17%	$\pm 0.19\%$	20

5-shot prototype classification reached 95.17%, 3.1 points below the supervised Swin-T result with 20 labels instead of 2,696.

3.3 PlantDoc, no adaptation

Model	PlantVillage Acc	PlantDoc Acc	Drop
ResNet50	97.92%	23.81%	74.1 pts
Swin-T	98.27%	39.42%	58.9 pts

Random chance on 3 classes is 33.33%. ResNet50's 23.81% is below this; the model is not random but consistently wrong. Swin-T retains more accuracy (39.42%), 15 points higher than ResNet50.

Per-class recall on PlantDoc:

Class	ResNet50	Swin-T
Gray Leaf Spot	50.7%	46.3%
Common Rust	3.4%	27.4%
Northern Leaf Blight	26.8%	44.3%

ResNet50 correctly identifies field rust 3.4% of the time despite scoring F1=1.000 on lab rust. Swin-T does eight times better on this class but is still well below in-domain performance.

24% of ResNet50's PlantDoc predictions were "Healthy" and 21% of Swin-T's were. PlantDoc contains no healthy corn images, so all of these were errors on diseased plants.

3.4 Few-shot on PlantDoc

The Swin-T embedding space had an inter/intra L2 distance ratio of 1.62 on PlantVillage but 1.03 on PlantDoc. On PlantDoc, same-class images are barely closer to each other than to different-class images.

Setting	Accuracy	95% CI	Labels
Zero-shot Swin-T	39.42%	—	0
3-way 1-shot, cosine + L2-norm	41.57%	$\pm 0.54\%$	3
3-way 5-shot, Euclidean	51.02%	$\pm 0.50\%$	15
Supervised Swin-T (PlantVillage)	98.27%	—	2,696

Five PlantDoc support images per class added 12 points over zero-shot. The result stays far below in-domain performance because the Swin-T embedding space, optimized for PlantVillage, does not separate PlantDoc classes well.

4. Discussion

PlantVillage results overstate what these models can do in the field. Both architectures fall by 59 to 74 points when tested on web-scraped images. PlantVillage performance alone is not a reliable indicator of deployment readiness.

The architecture comparison looks different in-domain than out-of-domain. On PlantVillage, ResNet50 and Swin-T differ by 0.5 points. On PlantDoc the gap is 15.6 points. The case for transformers over CNNs in this task is mainly about robustness to shift, not raw accuracy.

Few-shot prototype classification works well when source and target distributions match (95.17% with 20 labels). It does much less when they don't (51.02% with 15 PlantDoc labels). The bottleneck under shift is the embedding space, not the classifier. Chen et al. (2019) report similar findings; updating features, not just decision rules, is what closes severe gaps.

The healthy-prediction error rate matters separately from accuracy. A system that calls 22% of diseased plants healthy fails differently than one that confuses two diseases. For agricultural deployment, false negatives on disease should be reported alongside overall accuracy.

5. Conclusion

ResNet50 and Swin-T trained on PlantVillage both lose most of their accuracy on PlantDoc. Swin-T loses less. Few-shot prototype classification recovers in-domain performance with very few labels but does not bridge the cross-domain gap on its own. About one in five PlantDoc predictions from each model places diseased plants in the healthy class, an asymmetric error type that matters for any practical use.

Possible next steps include fine-tuning a fresh classifier head on K labeled PlantDoc images, parameter-efficient adapters such as LoRA on Swin-T's last block, self-supervised pretraining on field imagery, and a larger field test set covering more conditions.

References

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